

Muyang Zheng

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Education

University of California, Davis <i>Master of Science in Computer Science, GPA: 3.83/4.0</i>	California, USA Sept 2025 - May 2027
Hefei University of Technology <i>Bachelor of Engineering in Information Security</i>	Anhui, China Sept 2021 - June 2025

Publications

Open-ended Video Game Glitch Detection with Agentic Reasoning and Temporal Grounding (Accepted at <i>ACM Multimedia 2026</i>) [Preprint link] [Code link] <i>Muyang Zheng</i> , Tong Zhou, Geyang Wu, Zihao Lin, Haibo Wang, Lifu Huang*	Apr 2026
MIST: Jailbreaking Black-box Large Language Models via Iterative Semantic Tuning (Under review) [Preprint link] [Code link] <i>Muyang Zheng</i> , Yuanzhi Yao*, Changting Lin, Caihong Kai, Yanxiang Chen, Zhiquan Liu	June 2025

Professional Experience

Research Assistant , PLUM Lab at UC Davis – Advised by Dr. Lifu Huang	Sept 2025 – Present
- Built GLIDE, a LangGraph-based multimodal agent for open-ended video game glitch detection, with a five-stage pipeline for video scanning, iterative analysis, temporal grounding, and structured output generation. - Implemented a Planner–Executor–Reflector agent loop with VLM-based visual QA, targeted region zoom-in, and SAM3 object tracking; designed multi-agent debate and confidence-based stopping for iterative glitch verification. - Developed bidirectional temporal grounding to merge fragmented evidence and refine event boundaries, improving average F1 from 14.47% to 36.05% and mIoU from 0.28 to 0.51 across six multimodal backbones. - Built VIDEOGLITCHBENCH, a benchmark of 5,238 videos across 120 games, using an LLM-assisted annotation pipeline with human verification, glitch descriptions, and temporal annotations.	
Research Intern , Binjiang Institute of Zhejiang University – Advised by Dr. Meng Han	Aug 2024 – Apr 2025
- Benchmarked in-house LLM inference with BitNet and LLMPerf, analyzing throughput, latency, and 10,000+ API requests to identify performance bottlenecks. - Implemented 20+ jailbreak attacks, including GCG and AutoDAN, along with keyword-based refusal detection and LLM-as-judge evaluation for an LLM safety platform. - Built an automated red-teaming pipeline for commercial LLM APIs and ran 1,000+ jailbreak evaluations to analyze attack success rates and model robustness.	

Open-source Projects

Monocular Depth Estimation Model Fine-tuning PyTorch, LoRA [Project link]	Nov 2025
- Fine-tuned MiDaS DPT-Large on NYU Depth V2 using LoRA ($r=8$, $\alpha=16$), building a PyTorch training and evaluation pipeline with only 3.26M trainable parameters (0.94%). - Improved $\delta 1$ accuracy from 0.796 to 0.831, while reducing AbsRel from 0.155 to 0.135 and RMSE from 0.529 to 0.447 over the zero-shot baseline.	
LLM Jailbreak Evaluation Framework Python, Flask, Vue, SQLite [Project link]	May 2025
- Built a full-stack LLM red-teaming platform with Flask, Vue.js, and SQLite, integrating over 15 jailbreak attacks and automated evaluation across commercial LLM APIs. - Implemented MIST, a black-box jailbreak method based on iterative semantic optimization, and built automated attack analysis/visualization pipelines; the project was awarded Outstanding Graduation Project in Anhui Province.	

Skills

Coding Languages: Python, Java, C++ , SQL, JavaScript, C

ML/AI: PyTorch, Hugging Face Transformers, LoRA/PEFT, Vision-Language Models, Agentic AI, RAG, SAM3

LLM Systems: LangGraph, Agent Harness, vLLM, OpenAI-compatible APIs, LLM Evaluation & Red-Teaming

Software & Tools: Flask, React, Vue.js, SQLite, Git, Linux

Honors

Third-class Scholarship & Merit Student in 2021-2022 Academic Year

Outstanding Member of the Student Union in 2021-2022 Academic Year